

LOCUS

Unified Multi-Vendor DVR/NVR Forensic Analysis Suite

Problem Statement ID: 26150 (SIH 2026)

Target Organization: National Technical Research Organisation (NTRO)

Category: Software

Theme: Cybersecurity & Digital Forensics

Executive Summary

Locus is a next-generation, cross-platform desktop forensic application designed to solve the critical challenge of surveillance evidence fragmentation in law enforcement and intelligence operations.

Modern CCTV and DVR/NVR systems from major manufacturers—such as **Dahua, CP Plus, Honeywell, HIKVISION, TP-Link, Godrej, Uniview, and Matrix**—use proprietary file systems, raw disk structures, non-standard container headers, and custom video stream encodings. During criminal investigations, forensic officers are forced to juggle multiple vendor-specific playback utilities, struggle with timestamp drift across cameras, face extreme difficulty in recovering deleted footage, and risk evidence contamination.

Locus bridges this gap by offering a unified, vendor-agnostic software suite. It automates raw sector parsing, deleted video carving, multi-camera timeline synchronization, cryptographic evidence verification (SHA-256 & MD5), and computer vision AI analytics—delivering legally defensible, court-ready forensic reports from a single desktop interface.

Key Features & Core Capabilities

- **Automated OEM & File System Identification:** Detects proprietary partition layouts, magic header signatures, and video codecs automatically from raw disk dumps (.dd, .raw, .E01).
- **Sector-Level Video Carving & Deleted Recovery:** Bypasses corrupted or missing file system indexes by scanning raw drive sectors to carve deleted, overwritten, or damaged

H.264/H.265 video stream fragments.

- **Multi-Camera Synchronized Master Timeline:** Calibrates internal DVR clock drifts and synchronizes heterogeneous camera channels into a unified chronological master timeline with multi-grid web playback.
- **Cryptographic Chain-of-Custody Integrity:** Calculates and verifies SHA-256 and MD5 checksums at disk ingestion, file carving, extraction, and report export—ensuring strict legal admissibility (ISO/IEC 27037 compliant).
- **AI-Powered Computer Vision Analytics:** Integrated YOLOv8 models process extracted frames to index detected targets (*people, vehicles, objects*), facial bounding boxes, and motion heatmaps—enabling instant timeline filter queries.
- **Court-Ready Forensic Reporting:** Generates audit-compliant PDF/HTML reports featuring execution logs, evidence hash parity tables, carved frame thumbnails, and investigator signatures.

Technical Architecture & Tech Stack

Locus is built as a **monorepo standalone desktop application**, running a native Electron frontend wrapper powered by an embedded Python FastAPI engine.

Component	Technology Stack	Role & Purpose
Desktop Shell	Electron	Cross-platform desktop shell (Windows, Linux, macOS) with native OS file dialogs
Frontend UI	React + TypeScript + Vite	High-performance SPA with interactive video grid & timeline player
UI Styling	Tailwind CSS + ShadcnUI	Professional dark-mode design system matching enterprise forensic software
Single Backend	FastAPI (Python 3.11+)	Unified local API, raw sector carving, PyAV stream remuxing, WebSockets
Database	SQLite 3	Zero-configuration single-file database (forensics.db)
AI Engine	ONNX Runtime (YOLOv8)	Lightweight, hyper-fast local CPU/GPU object, face, and motion detection
Evidence Hashing	Python hashlib	Cryptographic SHA-256 and MD5 chain-of-custody verification

Team Roles & Responsibilities

Role #	Position	Primary Responsibilities	Core Stack
1	Systems Architect & Team Lead	Overall system design, database schemas, API contracts, SIH presentation strategy	FastAPI, Architecture, SQLite, Docker
2	Full-Stack Developer	Case intake UI, evidence management APIs, REST/WebSocket integration	React, TypeScript, FastAPI, SQLite
3	Python & Systems Forensics Dev	Binary disk image parsing, OEM header carving, PyAV/FFmpeg stream remuxing	Python, Binary I/O, PyAV, FFmpeg
4	AI / Computer Vision Dev	ONNX Runtime pipeline, YOLOv8 object/face detection, motion heatmaps	Python, ONNX, YOLOv8, OpenCV
5	Frontend & UI/UX Specialist	Multi-camera video sync player, interactive timeline matrix, report UI	React, TypeScript, Canvas, Video APIs
6	Docs, QA & Research Specialist	OEM comparative paper, SOPs, legal hash validation testing, final SIH slides	Technical Writing, QA, Research, SOPs

Key Differentiators

- **Cross-Platform Linux & Windows Compatibility:** While commercial tools are locked strictly to Windows, Locus runs natively on Linux (which NTRO and defense agencies prefer).
- **Zero-Configuration Desktop Executable:** Ships as a single .exe or .AppImage. No complex server setups or database installations required.
- **Integrated Computer Vision & Timeline Search:** Combines raw sector carving with instant natural language AI indexing ("*Find red car on Channel 2*").
- **Lightweight ONNX AI Execution:** Runs fast on standard laptop CPUs without requiring expensive \$2,000 gaming GPUs.